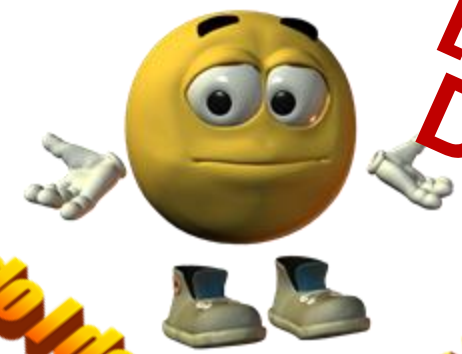




Seattle!
the city of coffee



**[ROUGH
DRAFT]**

What do I do?

so many amazing local roasters



Washington Coffee Matcher

Review data I scraped
from my coffee
subscription service

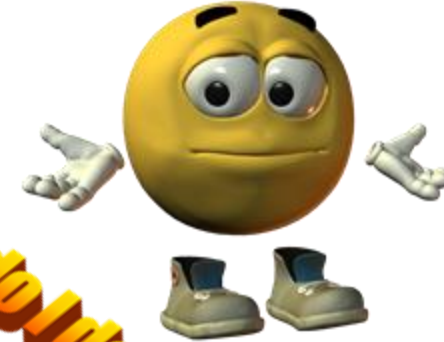
ft
&

Additional roaster information
we decide on (mission, store
availability, etc)

Background/Introduction

**[ROUGH
DRAFT]**

We scraped Washington coffee roaster data from a supplying subscription service and hoped to first supplement, and these “percolate” these layers into a suggestion tool, perhaps even adding local taste availability



What do I do?
so many amazing local roasters

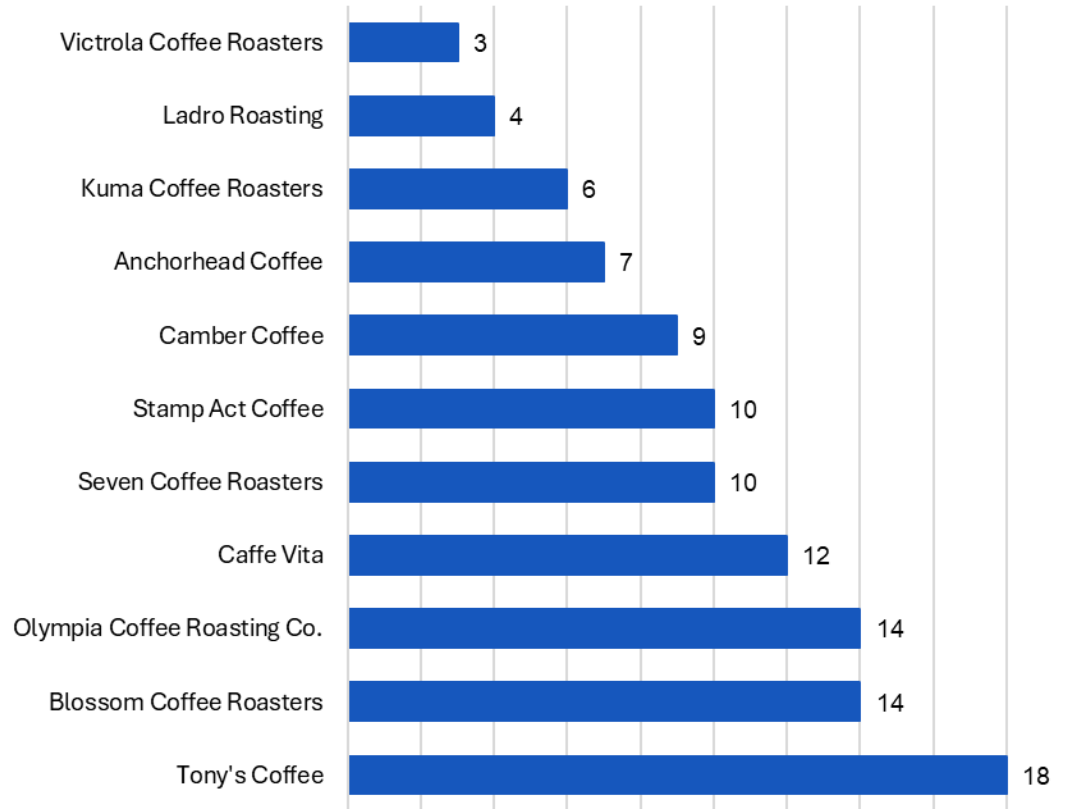
Washington Coffee Matcher

- Julia Glasser
- Johnny Liang
- Jeremy Wise
- Alvin Alias

Data

Washington has over 100 roasteries and our scraped dataset contains product, review, and rating information on over 100 of the coffees. Our scraped sampling of roasteries is shown below with the largest being "Tony's."

Our sample of Washington roasters and their number of coffee types (scraped data)

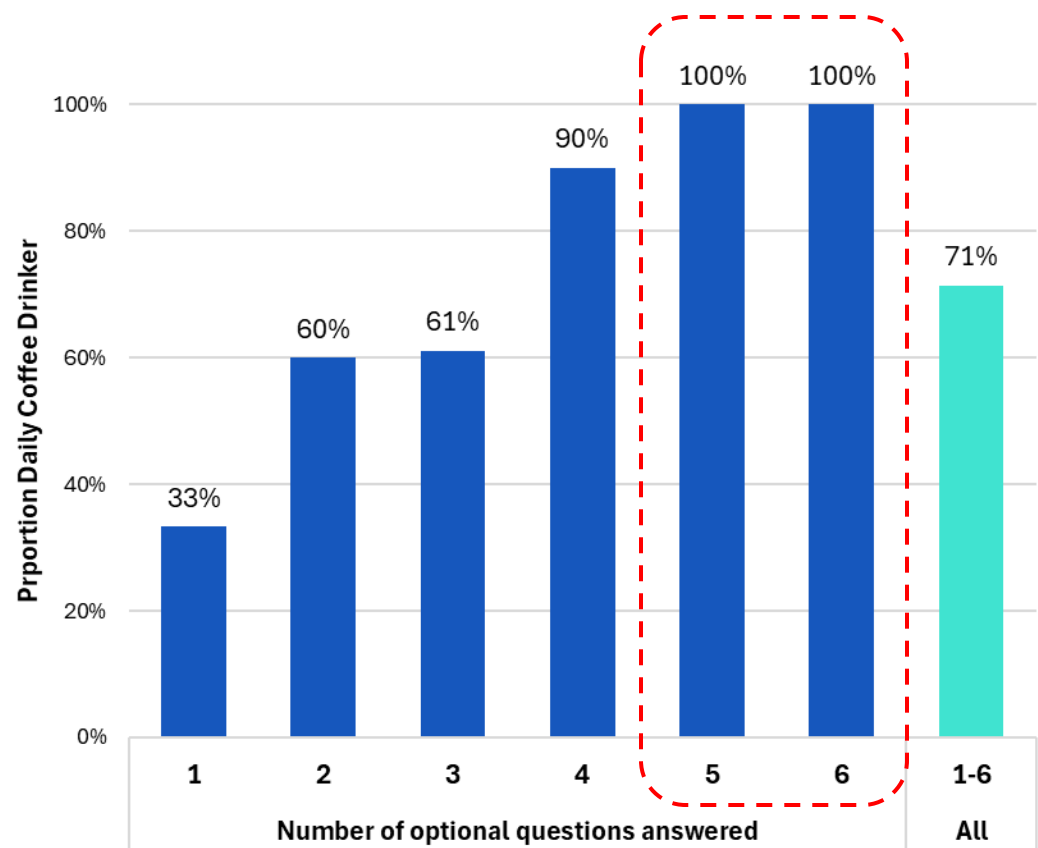


- Two primary datasets (scraped)
 - Products: Price, Shade, Origin, Rating, etc.
 - Reviews: Tasting notes
- Supporting Class Survey
- Supporting "Business Name" data search
 - King County Restaurant Inspection API
 - 60-70% match rate
 - ~~Washington Department of Revenue~~
 - Not machine searchable
 - ~~Kroger Product API~~
 - Low match rate anticipated

Survey

We tallied almost 50 surveys - mostly from class - and found that about 70% of drank one or more cups of coffee per day. Survey participants who elected to respond to more questions, were more likely to be daily coffee drinkers.

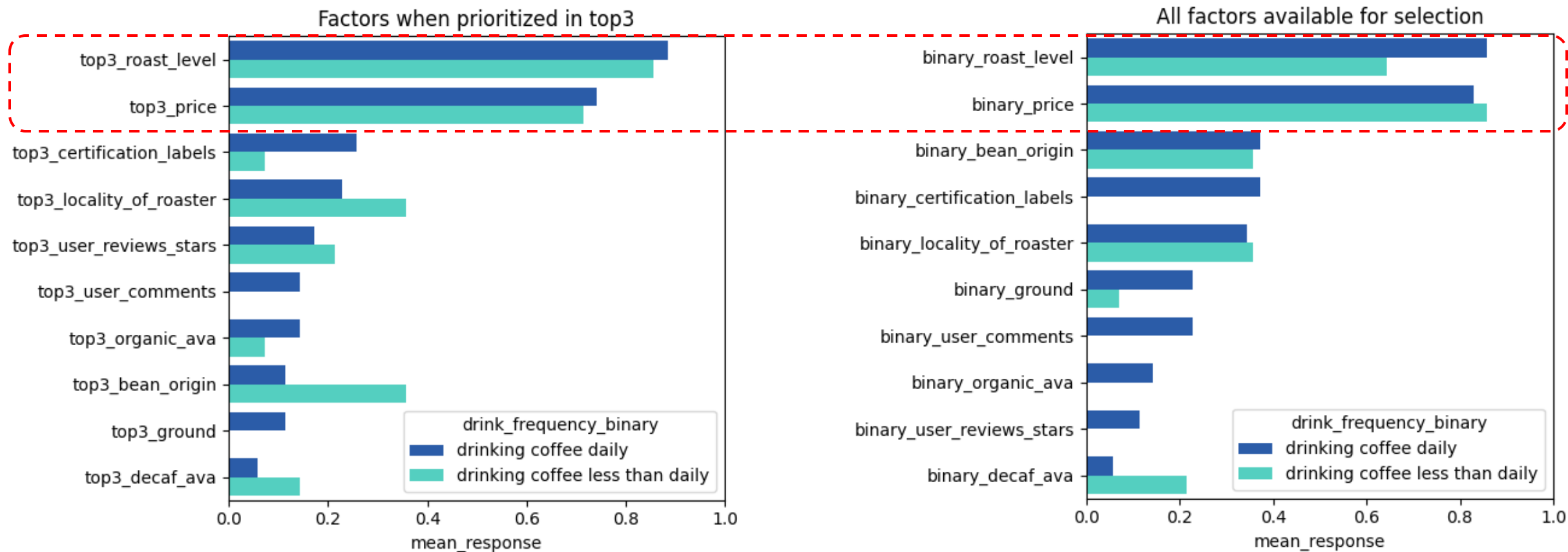
Optional questions responded to and liklihood of drinking one or more cups of coffee each day



- We appreciate the surveys
- Validates avid coffee consumer's interest
- But, , , data Lacks complete coffee information
- [please bear in mind the sequence which we presented the questions was not randomized]

Note:Please bear in mind the sequence which wepresented the questions was not randomized
Source: Survey

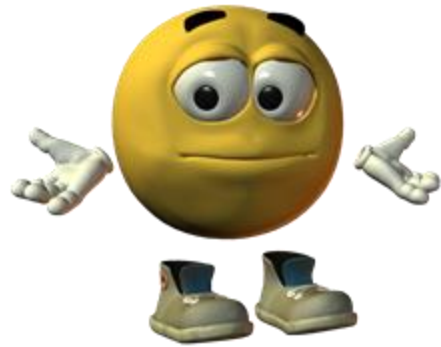
Regardless of consumption level, most survey participants prioritize price and roast level. However, preferences for other surveyed features varied more based on drinking frequency.



Users & Components

We see at least two retail clients using the tool: the person seeking the perfect match and the person seeking variety. But most users probably don't fit exclusively into a single grouping; roasters themselves may be users too.

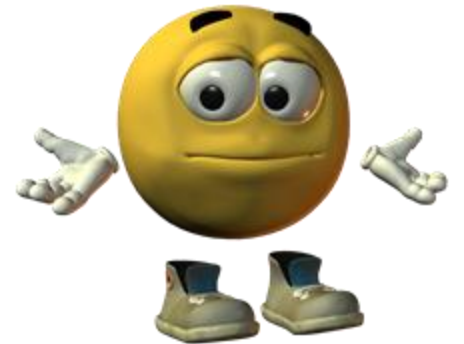
Single match?



What do I do?



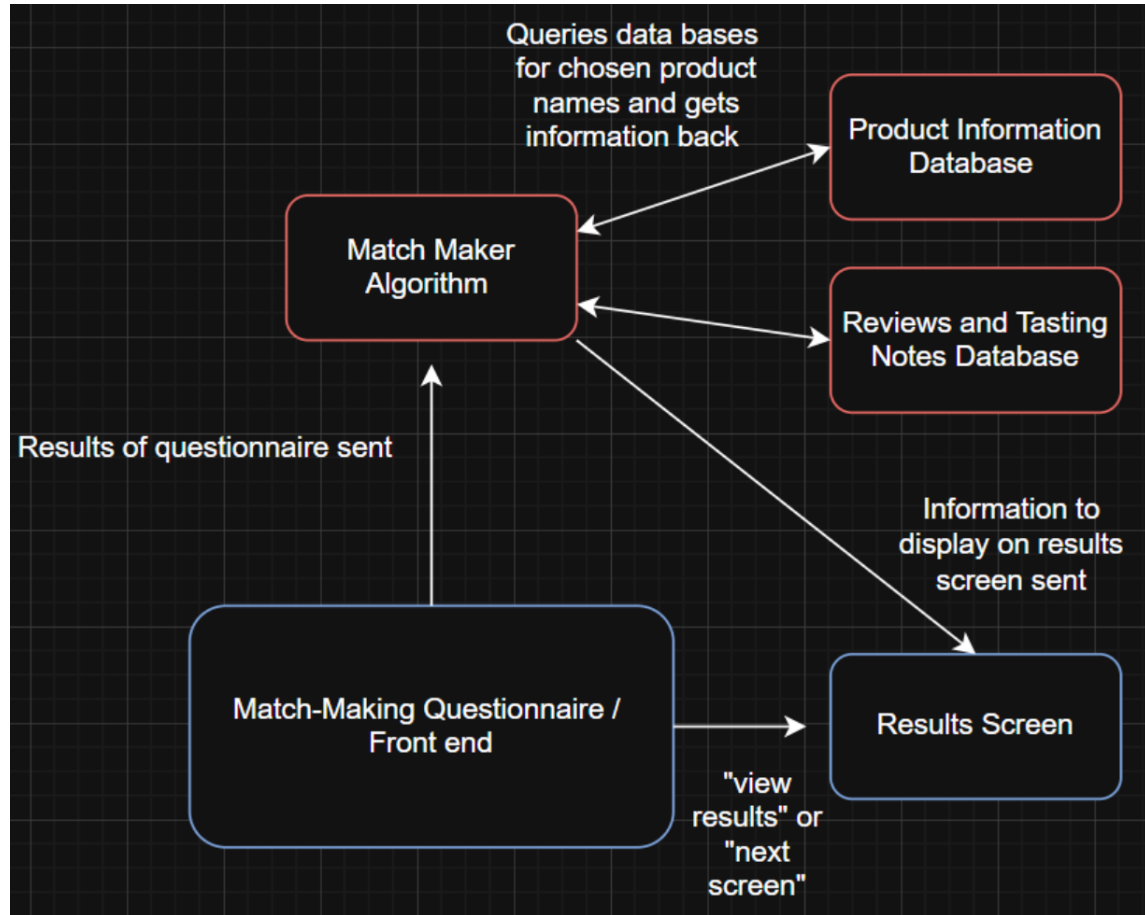
Multiple match?



So many amazing local roasters



Our component design caters primarily toward retail users. However, the object-oriented architecture helps facilitate the addition of roastery's assisted addition of new products into the product database.



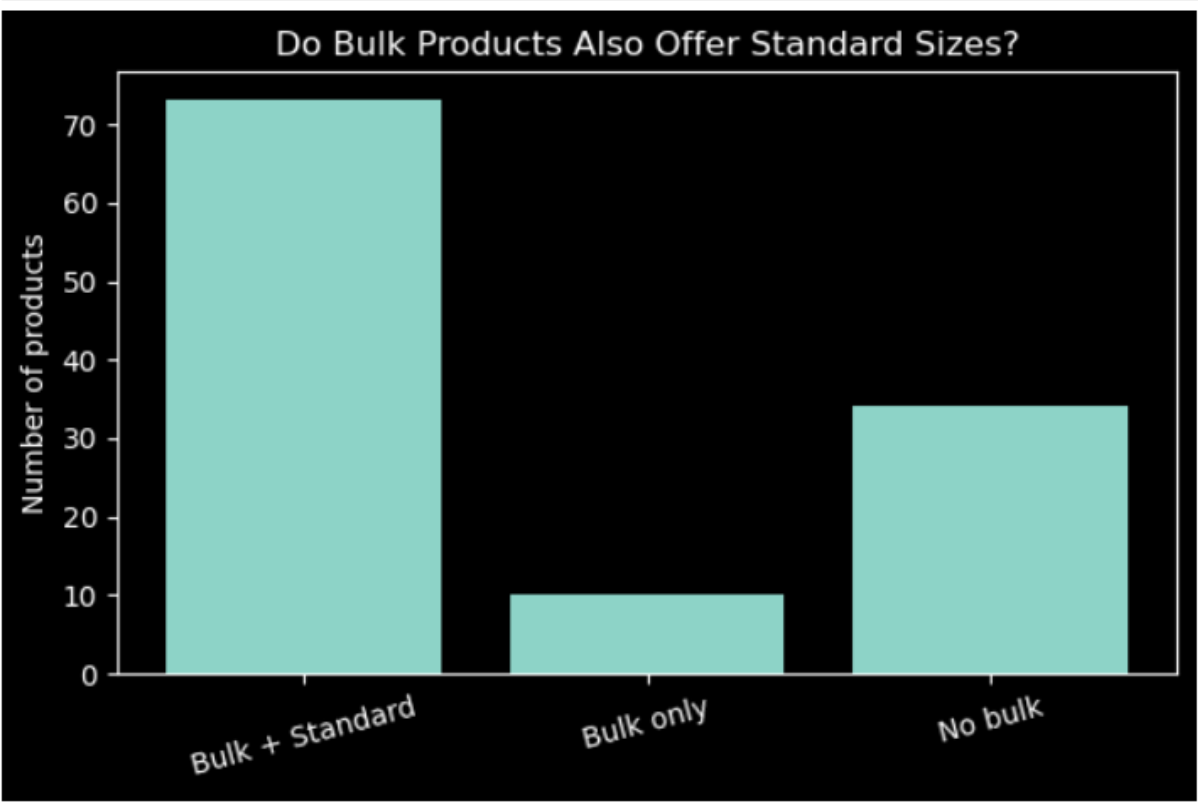
- Dictionary and/or OOP style eases communication between front and back end
- Roasteries product catalog constantly changing, ultimately tool provides easier access

Roaster Forms

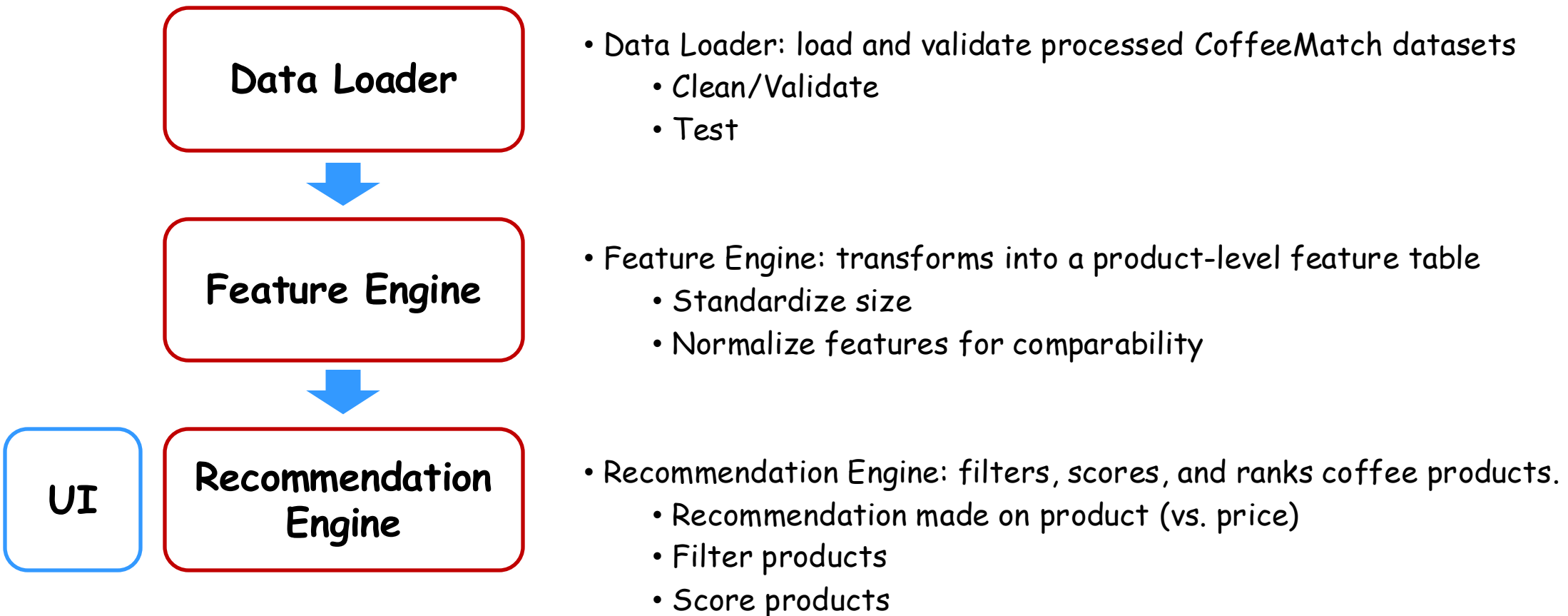
Design

- Feature Engineering
- Backend Pipeline
- Algorithm

About a dozen products were only sold in bulk. But we wanted to focus the app on small package (10-12 oz) pricing, so we needed to estimate pro forma pricing for the "bulk only" coffees from our scraped data.



Our backend architecture is divided into three primary components. It ultimately partners with the Streamlit UI to provide a recommendation.



The schemas serve as a shared contract between backend modules and the Streamlit UI. This allows the recommendation algorithm to filter, normalize, and weight user preferences in order to provide a scored product recommendation.

Class UserPreferences:

User selections from the UI (optional / **required**)

- `roast_type` : 'Light', 'Medium', or 'Dark'
- `max_price_per_oz` :
- `decaf` :
- `ground_required` :
- `single_origin_preference` :
- `blend_preference` :
- **`roast_weight`** : 0.45
- **`price_weight`** : 0.35
- **`popularity_weight`** : 0.20



Class Recommendation:

Recommendation(s) returned (optional / **required**)

- Similarities to UserPreferences, but adds:
 - **`score`**: float ~5.0
 - **`match_reasons`**: List[str]
 - `available_sizes`
 - `total_reviews`
 - `heart_percentage`
 - `has_reviews`
 - `url`

UI / Demo

The Streamlit landing page invites the user to filter - on the left - and weight on the right. From the survey, we found that roast level was important for regular and occasional coffee drinkers.

Coffee Match

 Find the Washington Bean of your Dreams 

Are you looking for a caffeinated or decaf coffee?

☒ Caffeinated! ⚡

☐ Decaf 😴

What's your roast preference?

☒ Light

☐ Medium

☐ Dark

☐ No preference / I'm not sure

Ground or whole beans (do you have a grinder)?

☒ Whole beans (yes)

☐ Pre-ground (no)

How important is roast for your match? (1 = least, 5 = most)

3



How important is ground status for your match? (1 = least, 5 = most)

2



Find your match!

The recommendation page is tailored to the user type and includes a final algorithm “match” score and other key coffee attributes, potentially including where to locally taste the coffee, if known.

MATCH #1	MATCH SCORE	PRICE PER OZ
Small Farms	5.00/5	\$0.96

Multiple match



MATCH #1	MATCH SCORE	PRICE PER OZ
Small Farms	5.00/5	\$0.96
MATCH #2	MATCH SCORE	PRICE PER OZ
Espresso Velluto Organic	4.94/5	\$1.00
MATCH #3	MATCH SCORE	PRICE PER OZ
Nectar	4.92/5	\$1.01



Single match

Café Location



Lessons/Next Steps

- More cafés and roasters, and more qualitative info
- More coffee entry database accessibility for roasters

Questions

Tests